ResQFlow-Flood

GitHub Suggestions - Foundation & Related Work

Information storage base for choosing an external GitHub project to extend with ResQFlow innovations.
Selection criteria: (1) already covers sensing/feedback gaps, (2) room to add Flood-GAPD / Ellipse / Polygon / Hybrid / 8-check verify / repair / multimodal vehicles, (3) runnable + clear license where possible.
Research date: 20 Aug 2026. ~25+ repositories investigated. Verify licenses and runability before forking.

1. Primary recommendation

Water Watch (best external base if you must fork)

URL: https://github.com/mishanilkazreen/water-watch

- License: MIT
- Stack: FastAPI + SQLite + React/Vite + Mapbox + OpenRouteService
- Weighted fit (judgment): ~4.55 / 5
- Already has: citizen + emergency hazard reports; flood/blocked-road hazards; shelters; UK Environment Agency flood data; hazard buffering into avoidance polygons; safe route recalculation
- Your innovation room: add Flood-GAPD, Ellipse/Polygon/Hybrid ranking, bus/truck/boat modes, 8-check verify, transactional repair, traces, job/vehicle assignment
- Caveats: Mapbox/ORS API keys; UK flood feed - wrap behind provider interface + Open-Meteo / fixtures for India demo

Practical result: keep Water Watch sensing + map shell; port ResQFlow Python dispatch modules behind it. Creates a clear before/after academic story: their sensing, your closed-loop innovation.

2. Important alternative: keep ResQFlow

Lowest-risk engineering decision remains: keep ResQFlow-Flood as the core and clean-room implement Water Watch-style report-to-hazard patterns (plus Open-Meteo, operator priority form). Only fork Water Watch if faculty requires starting from an external GitHub base.

3. Top licensed candidates (ranked)

A. Water Watch - PRIMARY

URL: https://github.com/mishanilkazreen/water-watch

- Best for: sensing + feedback already present; easy to add your algorithms
- Verdict: Best external foundation

B. LIFELINE - verification / audit donor

URL: https://github.com/annatchijova/lifeline

- License: Apache-2.0
- Stack: Python + SQLite + operations UI
- Has: typed reports, source/freshness, contradictions, human approval, verification graph, hash-chained audit
- Gap: weaker live flood routing / fleet dispatch
- Verdict: Best pattern donor for 8-check evidence and operator gate - not full product shell

C. ResQ Sri Lanka - mature flood product

URL: https://github.com/disaster-response-sl/resq

- License: Apache-2.0
- Has: 39 flood gauges, rainfall, SOS, relief camps, road hazards, route guidance, admin dashboards
- Stack: Node / web / mobile + Leaflet
- Gap: harder to port Python NetworkX dispatch innovations; logistics > algorithmic assignment
- Verdict: Best mature flood product shell if reviewers want gauges/SOS more than Python alignment

D. Intelligent Emergency Resource Scheduling Simulation Platform

URL: https://github.com/just1n9877/Intelligent-Emergency-Resource-Scheduling-Simulation-Platform

- License: MIT
- Stack: FastAPI + NetworkX + PostGIS + OR-Tools + Next.js
- Has: greedy/priority/Dijkstra/A*/CVRP dispatch, blocked-road scenarios, metrics, algorithm compare, Docker/tests
- Gap: NO citizen/weather/live sensing - fails the bottleneck you identified
- Verdict: Best algorithmic base IF grading is method-compare; not best for sensing story

E. ADMS (Ambulance Dispatch Management System)

URL: https://github.com/qppd/ambulance-dispatch-management-system

- License: Apache-2.0
- Stack: Flutter + Firebase + Cloud Functions
- Has: citizen GPS incidents, full lifecycle, auto nearest-unit dispatch, atomic claim-and-try-next, offline sync
- Gap: no flood graph / multimodal depth routing; Firebase rewrite cost high
- Verdict: Best live dispatch workflow donor; poor flood-algorithm host

F. RescuNet

URL: https://github.com/YoussefElebiary/RescuNet

- License: MIT
- Stack: FastAPI + OSMnx/NetworkX + React + C++ routing; heavy ML (YOLO/LSTM)
- Has: multi-vehicle assignment, urgency/group scoring, graph routing
- Gap: weak operational feedback loop; ML assets add burden
- Verdict: Best graph-routing component donor

4. Strong feature matches - DO NOT fork without license fix

These looked closest to ResQFlow features but currently lack a clear reusable license (or packaging is broken). Study concepts only unless owner grants written permission.

CrisisFlow

URL: https://github.com/h4ck3r0/CrisisFlow

- Closest sensing fit: citizen reports, Open-Meteo, flood sim, dispatch, road blocks, FastAPI/NetworkX
- Blockers: no license; missing model artifact can prevent clean boot

RouteOut

URL: https://github.com/Andrii-Samiilenko/RouteOut

- Hazard physics, A* OSM routing, dynamic reroute, weather client, WebSocket citizen GPS
- Blockers: no license; missing graph assets / incomplete packaging

Kairos / Pulsegrid

URL: https://github.com/DoSomethingGreat07/Kairos

- SOS priority, Hungarian assignment, Dijkstra + Yen backups, explainability, WebSockets
- Blockers: no LICENSE file; weak flood sensing; DB-heavy

Flood Rescue Coordination (Vietnam)

URL: https://github.com/longtq2501/Flood-Rescue-Coordination-and-Relief-Management-System

- Citizen GPS/photo requests, coordinator verify, team/vehicle assign, SSE feedback
- Blockers: no license; seven Java microservices - migration too heavy

Other no-license / high-risk (study only)

- GeoRescue Omni GIS: https://github.com/imanshadilshan/geo-rescue-omni-GIS-agent
- AegisFlowAI (GPL-3.0 - copyleft risk): https://github.com/ASEAN-AI-DZ/AegisFlowAI
- SEVA-AI (cloud-heavy / license unclear): https://github.com/koustubh-v/Seva-AI

5. Additional related-work repos (earlier scan)

- A-RESCUE (shelter matching, hurricane ABM): https://github.com/umnilab/A_RESCUE
- FIRM flood ABM: https://github.com/nclwater/FIRM
- LifeSim (USACE flood consequences): https://github.com/USACE-RMC/LifeSim
- Simulation-Assisted Evacuation Planning (MIP-LNS): https://github.com/KAI10/Simulation-Assisted-Evacuation-Planning
- DisasterEvacuationSystem (Dijkstra + shelter capacity): https://github.com/RohitDSonawane/DisasterEvacuationSystem
- equa-response (flood playbooks, Monte Carlo): https://github.com/Pawan-Prabhashana/equa-response
- arus Banjir Drill: https://github.com/SunflowersLwtech/arus
- ResQnet-AI: https://github.com/vergisodd/ResQnet-AI/
- AI Disaster Evacuation Planner: https://github.com/peelajanu/AI-Disaster-Evacuation-Planner
- ambulance-dispatch-systems (Dijkstra vs A*): https://github.com/Maxencejules/ambulance-dispatch-systems
- Crisis-Shield-AI (boats + ambulances): https://github.com/veerakarthick235/Crisis-Shield-AI
- AeroRescue (voice triage): https://github.com/swatiicfai/AeroRescue-
- City-Sync (citizen intake patterns): https://github.com/princ3kr/City-Sync
- Resgrid Core (mature CAD, ASP.NET): https://github.com/Resgrid/Core

6. Comparison snapshot (innovation room)

Water Watch - Fit high - Sensing yes - Feedback yes - Easy to add GAPD/geometry/checks - MIT

ResQ Sri Lanka - Fit high - Sensing yes - Feedback yes - Harder algorithm port - Apache-2.0

LIFELINE - Fit high for verify - Sensing typed reports - Feedback approvals - Add fleet/routing yourself - Apache-2.0

Emergency Scheduling Platform - Fit medium - Sensing no - Feedback scenario-only - Easiest algorithm insert - MIT

ADMS - Fit medium - Sensing citizen GPS - Feedback excellent - Flood algorithms costly - Apache-2.0

RescuNet - Fit medium - Sensing ML/text - Feedback weak ops - Good routing donor - MIT

CrisisFlow / RouteOut / Kairos - Feature-rich - Often no license - Do not fork as base

7. What your innovation should still own (on any base)

- Flood-GAPD prioritization of groups/jobs
- Weighted / Ellipse / Polygon / Hybrid candidate ranking
- Eight deterministic flood-evacuation verification checks
- Transactional repair (try next vehicle/route/shelter)
- Bus vs high-clearance vs boat depth constraints
- NetworkX evidence / replayable traces
- Closed-loop vs open-loop comparison (unsafe actuations)

8. Suggested migration boundary (if forking Water Watch)

Water Watch reports + map -> Report normalizer (trust, freshness, duplicates) -> Flood-GAPD -> Weighted/Ellipse/Polygon/Hybrid ranking -> 8-check verify + transactional repair -> Mission status feedback

9. Decision rule

- External fork required by faculty/mentor -> choose Water Watch (MIT)
- External fork NOT required -> keep ResQFlow-Flood; borrow licensed patterns only
- Need verification/audit inspiration -> study LIFELINE (Apache-2.0)
- Need algorithm-compare shell -> Emergency Scheduling Platform (MIT) + add sensing yourself
- Never copy CrisisFlow/RouteOut/Kairos/Vietnam Flood Rescue code without a license grant

10. Document control

Companion to docs/ResQFlow-Flood-Sensing-Feedback-Priorities.pdf

Canvas summary also exists: canvases/github-foundation-selection.canvas.tsx (Cursor IDE)

Regenerate this PDF by re-running the generation script if you update recommendations.